

4.4 Fractional Exponents and Radicals

LESSON FOCUS

Relate rational exponents and radicals.



Make Connections

Coffee, tea, and hot chocolate contain caffeine. The expression $100(0.87)^{\frac{1}{2}}$ represents the percent of caffeine left in your body $\frac{1}{2}$ h after you drink a caffeine beverage.

Given that $0.87^1 = 0.87$ and $0.87^0 = 1$, how can you estimate a value for $0.87^{\frac{1}{2}}$?

Construct Understanding

TRY THIS

Work with a partner.

- A.** Copy then complete each table. Use a calculator to complete the second column.

x	$x^{\frac{1}{2}}$
1	$1^{\frac{1}{2}} =$
4	$4^{\frac{1}{2}} =$
9	
16	
25	

x	$x^{\frac{1}{3}}$
1	
8	
27	
64	
125	

Continue the pattern. Write the next 3 lines in each table.

B. For each table:

- What do you notice about the numbers in the first column?
Compare the numbers in the first and second columns.
What conclusions can you make?
- What do you think the exponent $\frac{1}{2}$ means? Confirm your prediction by trying other examples on a calculator.
- What do you think the exponent $\frac{1}{3}$ means? Confirm your prediction by trying other examples on a calculator.

C. What do you think $a^{\frac{1}{4}}$ and $a^{\frac{1}{5}}$ mean?

Use a calculator to test your predictions for different values of a .

D. What does $a^{\frac{1}{n}}$ mean? Explain your reasoning.

In grade 9, you learned that for powers with integral bases and whole number exponents:

$$a^m \cdot a^n = a^{m+n}$$

We can extend this law to powers with fractional exponents with numerator 1:

$$\begin{aligned} 5^{\frac{1}{2}} \cdot 5^{\frac{1}{2}} &= 5^{\frac{1}{2} + \frac{1}{2}} & \text{and} & \quad \sqrt{5} \cdot \sqrt{5} = \sqrt{25} \\ &= 5^1 & & \quad = 5 \\ &= 5 \end{aligned}$$

$5^{\frac{1}{2}}$ and $\sqrt{5}$ are equivalent expressions; that is, $5^{\frac{1}{2}} = \sqrt{5}$.

$$\begin{aligned} \text{Similarly, } 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} &= 5^{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} & \text{and} & \quad \sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = \sqrt[3]{125} \\ &= 5^1 & & \quad = 5 \\ &= 5 \end{aligned}$$

$$5^{\frac{1}{3}} = \sqrt[3]{5}$$

These examples indicate that:

- Raising a number to the exponent $\frac{1}{2}$ is equivalent to taking the square root of the number.
- Raising a number to the exponent $\frac{1}{3}$ is equivalent to taking the cube root of the number, and so on.

To multiply powers with the same base, add the exponents.

Powers with Rational Exponents with Numerator 1

When n is a natural number and x is a rational number, $x^{\frac{1}{n}} = \sqrt[n]{x}$

Example 1

Evaluating Powers of the Form $a^{\frac{1}{n}}$

Evaluate each power without using a calculator.

a) $27^{\frac{1}{3}}$ b) $0.49^{\frac{1}{2}}$ c) $(-64)^{\frac{1}{3}}$ d) $\left(\frac{4}{9}\right)^{\frac{1}{2}}$

SOLUTION

The denominator of the exponent is the index of the radical.

a) $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$ b) $0.49^{\frac{1}{2}} = \sqrt{0.49} = 0.7$

c) $(-64)^{\frac{1}{3}} = \sqrt[3]{-64} = -4$ d) $\left(\frac{4}{9}\right)^{\frac{1}{2}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$

CHECK YOUR UNDERSTANDING

1. Evaluate each power without using a calculator.

a) $1000^{\frac{1}{3}}$ b) $0.25^{\frac{1}{2}}$

c) $(-8)^{\frac{1}{3}}$ d) $\left(\frac{16}{81}\right)^{\frac{1}{4}}$

[Answers: a) 10 b) 0.5

c) -2 d) $\frac{2}{3}$]

How could you check your answers?

A fraction can be written as a terminating or repeating decimal, so we can interpret powers with decimal exponents; for example, $0.2 = \frac{1}{5}$, so $32^{0.2} = 32^{\frac{1}{5}}$.

We can evaluate $32^{\frac{1}{5}}$ and $32^{0.2}$ on a calculator to show that both expressions have the same value.

Why do we use brackets to evaluate when the exponent is a fraction, but not when the exponent is a decimal?

$32^{(1/5)}$

$\frac{2}{3}$

$32^{0.2}$

$\frac{2}{3}$

To give meaning to a power such as $8^{\frac{2}{3}}$, we extend the exponent law $(a^m)^n = a^{mn}$ so that it applies when m and n are rational numbers.

We write the exponent $\frac{2}{3}$ as $\frac{1}{3} \cdot 2$, or as $2 \cdot \frac{1}{3}$.

$$\begin{aligned} \text{So, } 8^{\frac{2}{3}} &= 8^{\frac{1}{3} \cdot 2} \\ &= \left(8^{\frac{1}{3}}\right)^2 \\ &= \left(\sqrt[3]{8}\right)^2 \end{aligned}$$

Take the cube root of 8, then square the result.

$$\begin{aligned} \text{So, } 8^{\frac{2}{3}} &= 2^2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} 8^{\frac{2}{3}} &= 8^{2 \cdot \frac{1}{3}} \\ &= (8^2)^{\frac{1}{3}} \\ &= \sqrt[3]{8^2} \end{aligned}$$

Square 8, then take the cube root of the result.

$$\begin{aligned} 8^{\frac{2}{3}} &= \sqrt[3]{64} \\ &= 4 \end{aligned}$$

These examples illustrate that the numerator of a fractional exponent represents a power and the denominator represents a root. The root and power can be evaluated in any order.

Powers with Rational Exponents

When m and n are natural numbers, and x is a rational number,

$$x^{\frac{m}{n}} = \left(x^{\frac{1}{n}}\right)^m \quad \text{and} \quad x^{\frac{m}{n}} = \left(x^m\right)^{\frac{1}{n}}$$

$$= \left(\sqrt[n]{x}\right)^m \quad \quad \quad = \sqrt[n]{x^m}$$

Example 2 Rewriting Powers in Radical and Exponent Form

- a) Write $40^{\frac{2}{3}}$ in radical form in 2 ways.
 b) Write $\sqrt{3^5}$ and $(\sqrt[3]{25})^2$ in exponent form.

SOLUTION

- a) Use $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$ or $\sqrt[n]{a^m}$.

$$40^{\frac{2}{3}} = (\sqrt[3]{40})^2 \text{ or } \sqrt[3]{40^2}$$

- b) Use $\sqrt[n]{a^m} = a^{\frac{m}{n}}$.

$$\sqrt{3^5} = 3^{\frac{5}{2}}$$

$$\text{Use } (\sqrt[n]{a})^m = a^{\frac{m}{n}}.$$

$$(\sqrt[3]{25})^2 = 25^{\frac{2}{3}}$$

The index of the radical is 2.

CHECK YOUR UNDERSTANDING

2. a) Write $26^{\frac{2}{5}}$ in radical form in 2 ways.

- b) Write $\sqrt{6^5}$ and $(\sqrt[4]{19})^3$ in exponent form.

[Answers: a) $(\sqrt[5]{26})^2$ or $\sqrt[5]{26^2}$

b) $6^{\frac{5}{2}}$, $19^{\frac{3}{4}}$]

Example 3 Evaluating Powers with Rational Exponents and Rational Bases

Evaluate.

a) $0.04^{\frac{3}{2}}$

b) $27^{\frac{4}{3}}$

c) $(-32)^{0.4}$

d) $1.8^{1.4}$

SOLUTION

a) $0.04^{\frac{3}{2}} = \left(0.04^{\frac{1}{2}}\right)^3$

$$= (\sqrt{0.04})^3$$

$$= 0.2^3$$

$$= 0.008$$

b) $27^{\frac{4}{3}} = \left(27^{\frac{1}{3}}\right)^4$

$$= (\sqrt[3]{27})^4$$

$$= 3^4$$

$$= 81$$

(Solution continues.)

CHECK YOUR UNDERSTANDING

3. Evaluate.

a) $0.01^{\frac{3}{2}}$

b) $(-27)^{\frac{4}{3}}$

c) $81^{\frac{3}{4}}$

d) $0.75^{1.2}$

[Answers: a) 0.001 b) 81

c) 27 d) 0.7080...]

c) The exponent $0.4 = \frac{4}{10}$ or $\frac{2}{5}$

$$\begin{aligned}\text{So, } (-32)^{0.4} &= (-32)^{\frac{2}{5}} \\ &= \left[(-32)^{\frac{1}{5}}\right]^2 \\ &= \left(\sqrt[5]{-32}\right)^2 \\ &= (-2)^2 \\ &= 4\end{aligned}$$

d) $1.8^{1.4}$

Use a calculator.

$$1.8^{1.4} = 2.277096874$$

$$1.8^{1.4} = 2.2770\dots$$

The powers in parts a to c were evaluated by taking the root first. Evaluate each power by raising the base to the exponent first. Which strategy is more efficient? Justify your answer.

Example 4 Applying Rational Exponents

Biologists use the formula $b = 0.01m^{\frac{2}{3}}$ to estimate the brain mass, b kilograms, of a mammal with body mass m kilograms. Estimate the brain mass of each animal.

- a) a husky with a body mass of 27 kg
- b) a polar bear with a body mass of 200 kg

SOLUTION

Use the formula $b = 0.01m^{\frac{2}{3}}$.

a) Substitute: $m = 27$

$$\begin{aligned}b &= 0.01(27)^{\frac{2}{3}} \\ b &= 0.01\left(\sqrt[3]{27}\right)^2 \\ b &= 0.01(3)^2 \\ b &= 0.01(9) \\ b &= 0.09\end{aligned}$$

Use the order of operations.
Evaluate the power first.

The brain mass of the husky is approximately 0.09 kg.

b) Substitute: $m = 200$

$$b = 0.01(200)^{\frac{2}{3}}$$

Use a calculator.

$$0.01(200)^{(2/3)} = 0.341995189$$

The brain mass of the polar bear is approximately 0.34 kg.

CHECK YOUR UNDERSTANDING

4. Use the formula $b = 0.01m^{\frac{2}{3}}$ to estimate the brain mass of each animal.
- a) a moose with a body mass of 512 kg
 - b) a cat with a body mass of 5 kg

[Answers: a) approximately 0.64 kg
b) approximately 0.03 kg]

Why did we use mental math to evaluate part a, but a calculator to evaluate part b?

Discuss the Ideas

- When a is a rational number and n is a natural number, what does $a^{\frac{1}{n}}$ represent?
- When a is a rational number and m and n are natural numbers, what does $a^{\frac{m}{n}}$ represent?

Exercises

A

- Evaluate each power without using a calculator.
 - $16^{\frac{1}{2}}$
 - $36^{\frac{1}{2}}$
 - $64^{\frac{1}{3}}$
 - $32^{\frac{1}{5}}$
 - $(-27)^{\frac{1}{3}}$
 - $(-1000)^{\frac{1}{3}}$
- Evaluate each power without using a calculator.
 - $100^{0.5}$
 - $81^{0.25}$
 - $1024^{0.2}$
 - $(-32)^{0.2}$
- Write each power as a radical.
 - $36^{\frac{1}{3}}$
 - $48^{\frac{1}{2}}$
 - $(-30)^{\frac{1}{5}}$
- Write each radical as a power.
 - $\sqrt{39}$
 - $\sqrt[4]{90}$
 - $\sqrt[3]{29}$
 - $\sqrt[5]{100}$
- Evaluate each power without using a calculator.
 - 8^0
 - $8^{\frac{1}{3}}$
 - $8^{\frac{2}{3}}$
 - $8^{\frac{3}{3}}$
 - $8^{\frac{4}{3}}$
 - $8^{\frac{5}{3}}$

B

- Write each power as a radical.
 - $4^{\frac{2}{3}}$
 - $(-10)^{\frac{3}{5}}$
 - $2.3^{\frac{3}{2}}$
- A cube has a volume of 350 cm^3 . Write the edge length of the cube as a radical and as a power.
- Write each power as a radical.
 - $48^{\frac{2}{3}}$
 - $(-1.8)^{\frac{5}{3}}$
 - $\left(\frac{3}{8}\right)^{2.5}$
 - $0.75^{0.75}$
 - $\left(-\frac{5}{9}\right)^{\frac{2}{5}}$
 - $1.25^{1.5}$

- Write each radical as a power.

- $\sqrt{3.8^3}$
- $(\sqrt[3]{-1.5})^2$
- $\sqrt[4]{\left(\frac{9}{5}\right)^5}$
- $\sqrt[3]{\left(\frac{3}{8}\right)^4}$
- $\left(\sqrt{\frac{5}{4}}\right)^3$
- $\sqrt[5]{(-2.5)^3}$

- Evaluate each power without using a calculator.

- $9^{\frac{3}{2}}$
- $\left(\frac{27}{8}\right)^{\frac{2}{3}}$
- $(-27)^{\frac{3}{2}}$
- $0.36^{1.5}$
- $(-64)^{\frac{2}{3}}$
- $\left(\frac{4}{25}\right)^{\frac{3}{2}}$

- Write an equivalent form for each number using a power with exponent $\frac{1}{2}$, then write the answer as a radical.

- 2
- 4
- 10
- 3
- 5

- Write an equivalent form for each number using a power with exponent $\frac{1}{3}$, then write the answer as a radical.

- 1
- 2
- 3
- 4
- 4

- Arrange these numbers in order from least to greatest. Describe your strategy.

$$\sqrt[3]{4}, 4^{\frac{3}{2}}, 4^2, \left(\frac{1}{4}\right)^{\frac{3}{2}}$$

- Evaluate.

- $16^{1.5}$
- $81^{0.75}$
- $(-32)^{0.8}$
- $35^{0.5}$
- $1.21^{1.5}$
- $\left(\frac{3}{4}\right)^{0.6}$

- Which powers in part a could you have evaluated without a calculator? How can you tell before you evaluate?

17. The height, h metres, of a certain species of fir tree can be estimated from the formula $h = 35d^{\frac{2}{3}}$, where d metres is the diameter at the base. Use the formula to determine the approximate height of a fir tree with base diameter 3.2 m.

18. Here is a student's solution for evaluating a power.

$$\begin{aligned} 1.96^{\frac{3}{2}} &= (\sqrt[3]{1.96})^2 \\ &= (1.2514\dots)^2 \\ &= 1.5661\dots \end{aligned}$$

Identify the errors the student made. Write a correct solution.

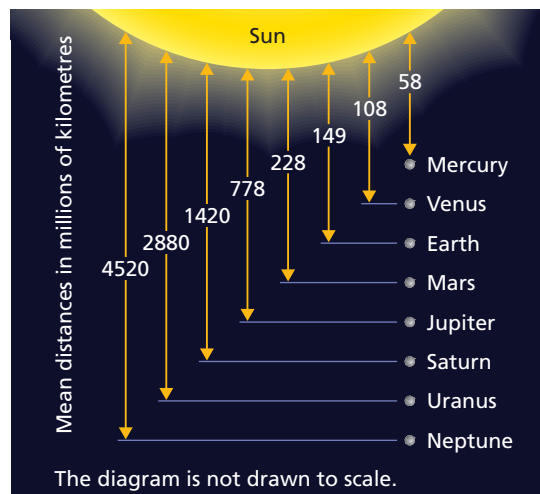
19. A formula for the approximate surface area, SA square metres, of a person's body is $SA = 0.096m^{0.7}$, where m is the person's mass in kilograms. Calculate the surface area of a child with mass 40 kg.

20. Here is an expression for the percent of caffeine that remains in your body n hours after you drink a caffeine beverage:

$$100(0.5)^{\frac{n}{5}}$$

- Show that this expression and the expression on page 222 give the same result, to the nearest whole number, for the percent of caffeine that remains after $\frac{1}{2}$ h.
- Use the expression above to determine the percent of caffeine that remains after 1.5 h.
- After how many hours does 50% of the caffeine remain? Explain how you know.

21. In the late 1500s, Johannes Kepler developed a formula to calculate the time it takes each planet to orbit the sun (called the *period*). The formula is $T \doteq 0.2R^{\frac{3}{2}}$, where T is the period in Earth days and R is the mean distance from the planet to the sun in millions of kilometres.



The mean distance of Earth from the sun is about 149 million kilometres. The mean distance of Mars from the sun is about 228 million kilometres. Which planet has the longer period, Earth or Mars? Justify your answer.

C

22. Two students discussed the meaning of the statement $3.2^{4.2} = 132.3213\dots$
- Luc said: It means 3.2 multiplied by itself 4.2 times is about 132.3213.
- Karen said: No, you can't multiply a number 4.2 times. $3.2^{4.2}$ can be written as $3.2^{\frac{42}{10}}$. So the statement means that 42 factors, each equal to the tenth root of 3.2, multiplied together will equal about 132.3213.
- Which student is correct? Explain.

Reflect

In the power $x^{\frac{m}{n}}$, m and n are natural numbers and x is a rational number. What does the numerator m represent? What does the denominator n represent? Use an example to explain your answer.

What must be true about x for $x^{\frac{m}{n}}$ to be a rational number?